Exploring the Linkage Between Energy Consumption and Economic Growth in BRICS Countries Through Disaggregated Analysis

Overview

- The main objective of this study is to analyze the impact of disaggregated energy consumption (coal, natural gas, petrol, and electricity) on the economic growth of BRICS countries (Brazil, Russia, India, China, and South Africa) from 1990 to 2020.
- The study implements the **Augmented Mean Group (AMG) and Cross-Sectional Augmented Autoregressive Distributed Lag (CS-ARDL)** techniques for empirical analysis. The cointegration results indicate a consistent long-term link between coal consumption, gas, petrol, electricity, and economic growth.
- The CS-ARDL estimates show that disaggregated energy consumption has a positive short and long-term effect on economic growth, and the AMG approach supports these findings.
- The **panel Granger** causality test result confirms the causal link between coal consumption and economic growth, electricity and economic growth, and petroleum and economic growth, supporting the feedback hypotheses,

Literature Review

- Existing empirical studies reveal four perspectives on the causal relationship between energy use and economic progress.
- The first perspective is that energy usage Granger causes economic progress, known as the “growth hypothesis.”
- The second perspective suggests that economic growth Granger causes energy usage, known as the “conservation hypothesis.”
- The third perspective proposes that both economic growth and energy usage have bidirectional causality, known as the “feedback hypothesis” or bidirectional causality.

- The fourth perspective indicates no causality between economic development and energy usage, called the “neutrality hypothesis,”

> Petroleum Consumption-Economic Growth

- Several earlier research studies have examined the causal link between petroleum (oil) usage and economic development. For example, Aqeel and Butt (2001) study shows that growth leads to an expansion in petroleum usage in Pakistan from 1995 to 1996.
- Lee and Chang (2005) study indicate a one-way causation from oil use to GDP in Taiwan from 1954 to 2003.
- Asghar (2008) assessed

the causal linkage between different sources of energy usage and economic progress for a sample of 5 South Asian nations, namely, Pakistan, Sri Lanka, India, Nepal, and Bangladesh.

Method

- The selected variables are important due to their widespread usage. Furthermore, the analysis covers the period from 1990 to 2020 due to data availability.

Table 1 Variables description and data sources

Variables	Description	Label
Economic growth	GDP (constant 2015 US$)	$gdp_{(i,t)}$
Electric power use (consumption)	KWh per capita	$electric_{(i,t)}$
Coal use	Millions of tons of oil equivalent	$coal_{(i,t)}$
Gas use	Millions of tons of oil equivalent	$gas_{(i,t)}$
Petrol use	Millions of tons of oil equivalent	$petrol_{(i,t)}$
Labor force	Total labor force	$labfr_{(i,t)}$
Foreign direct investment	Total FDI net-inflows	$fdi_{(i,t)}$

Source: Authors’ compilation

- The multivariate regression model within the economic growth theory framework by Solow (1956) and Mankiw et al. (1992) and also used in many prior studies

including Wang et al. (2023a), Azam (2022), Ali et al. (2022), Azam and Khan (2022), andAydin (2018) can be written as follows.

gdp = $\alpha_0 + \alpha_1 coal_{i,t} + \alpha_2 gas_{i,t} + \alpha_3 electric_{i,t} + \alpha_4 petrol_{i,t} + \alpha_5 fdi_{i,t} + \alpha_6 labfr_{i,t} + \epsilon i, t.....$

Estimation Procedure

- possible. The study proceeds with **Pesaran** (2007) cross-sectional dependence (CSD) test. The CSD test is important to ensure unbiased estimates and estimator efficiency. We further applied the **Covariate Augmented Dickey-Fuller (CADF)** and Cross-sectionally augmented Im, **Pesaran and Shin (CIPS)** tests, to ascertain the stationarity properties of the variables. However, once variables are stationary, a cointegration test is required to determine the possibility of a long-run association.

- However, once variables are stationary, a cointegration test is required to determine the possibility of a long-run association. In this contribution, the **Westerlund (2007) test** investigates the existence of cointegration.

- Moreover, the traditional techniques provide partial output, leading to wrong policy implication suggestions. Therefore, this research analyzed the long- and short-run output analysis using Chudik and Pesaran (2015) CD-ARDL. The mathematical form is given as follows:

- $R_{it} = \sum_{l=0}^{v0} \alpha_{li} R_{i,t-l} + \sum_{l=0}^{vl} \delta_{li} J_{i,t-l} + \epsilon_{it}$

- The problem with Eq. (4) is that it cannot solve the problems of the existence of a unit root, unobservable factors, slope-homogeneity, and cross-sectional-dependence. Therefore, modifying the above equation leads to the following:

$$R_{it} = \sum_{I=0}^{vo} \alpha_{Ii}, R_{i,t-1} + \sum_{I=0}^{vl} \delta_{Ii}, J_{i,t-1} + \sum_{I=0}^{vx} X\prime_{Ii}, \overline{IE}_{i,t-1} + \varepsilon_{it} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$

- Using cross-section averages, Eq. (5) resolves the concerns in the above discussion and eliminates the threshold effect (Chudik & Pesaran, 2015).

$$\overline{E}_{i,t-1} = (\overline{R}_{i,t-1}, \overline{J}_{i,t-1})$$

Equation (5) shows the averages and lags, while Eq. (6) shows the long-run oefficients.

$$\hat{\alpha}_{c.d-ARDL,i} = \left(\frac{\sum_{I=0}^{vl} \hat{\delta}_{Ii}}{1 - \sum_{I=0}^{vo} \hat{\alpha}_{Ii}} \right) \ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (6)$$

The mean group is given in Eq. (7) as below:

$$\hat{\overline{\alpha}}_{mg} = \sum_{i=1}^{n} \hat{\alpha}_i \ldots\ldots\ldots\ldots\ldots\ldots \quad (7)$$

- The short- and long-run coefficient analysis is calculated in Eq. (8) and given as follows:

$$\Delta R_{it} = \alpha_i \{R_{i,t-1} - \vartheta_i J_{it}\} - \sum_{I=0}^{vo-1} \alpha_{Ii}, \Delta_I R_{i,t-1} + \sum_{I=0}^{vl} \delta_{Ii}, \Delta_I J_{i,t-1} + \sum_{I=0}^{vx} X\prime_i, I\overline{E}_{i,t-1} + \varepsilon_{it} \ldots\ldots\ldots\ldots\ldots\ldots \quad (8)$$

$$\hat{\omega}_i = -\left(1 - \sum_{I=1}^{vR} \hat{\alpha}_{Ii}\right) \ldots\ldots\ldots\ldots\ldots\ldots \quad (8a)$$

Results

- Descriptive Statistics:

Table 2 Descriptive statistics of all variables

Variable	Mean	Std. Dev	Min	Max	Skewness	Kurtosis
gdp_{it}	3.6531	1.2183	1.2981	6.3175	0.3821	0.4724
$coal_{it}$	1.8316	0.3862	1.1983	3.7072	0.4728	0.5291
gas_{it}	2.5491	0.2438	0.7395	5.2746	0.5753	0.4962
$electric_{it}$	3.9016	1.0113	0.7551	4.6482	0.2842	0.5568
$petrol_{it}$	2.7883	0.6637	1.7137	3.2318	0.5326	0.4829
fdi_{it}	0.2759	0.1326	0.1084	2.3225	0.3996	0.6987
$labfr_{it}$	0.3152	0.0287	0.2581	0.9783	0.5826	0.4989

Source: Authors' computation

- Cross-Sectional Dependence test analysis

Variables	CDSTAT	Mean absolute value
$gdp_{(i,t)}$	15.410	0.905
$coal_{(i,t)}$	3.489	0.854
$gas_{(i,t)}$	13.324	0.782
$electric_{(i,t)}$	10.103	0.593
$petrol_{(i,t)}$	6.293	0.785
$fdi_{(i,t)}$	14.06	0.813
$labfr_{(i,t)}$	13.97	0.805

Source: Authors' computation

the results of the cross-sectional dependence test, which confirms the dependency among variables. The null hypothesis of no cross-section independence is rejected. The high mean absolute value indicates high dependence, confirm- ing that shocks in one BRICS country will affect the others.

- Slope Homogenity Test analysis

Model	stat	stat	*p*-value
$gdp_{(i,t)}coal_{(i,t)}gas_{(i,t)}electric_{(i,t)}petrol_{(i,t)}fdi_{(i,t)}labfr_{(i,t)}$	Tide	10.917 ***	0.000
$gdp_{(i,t)}coal_{(i,t)}gas_{(i,t)}electric_{(i,t)}petrol_{(i,t)}fdi_{(i,t)}labfr_{(i,t)}$	Delta tide	12.881 ***	0.000

Source: Authors' computation

the results of the slope homogeneity test, with low p-values rejecting the null hypothesis of slope homogeneity. These results confirm the hetero-geneity in cross-sectional coefficients.

- Unit root test analysis

Table 5 Unit root test analysis

Variables	CADF		CIPS	
	Level I (0)	First Diff; I (1)	Level I (0)	First Diff; I (1)
$gdp_{(i,t)}$	−1.816	−2.216***	−1.595	−2.502***
$coal_{(i,t)}$	−1.414	−2.815***	−1.843	−2.806**
$gas_{(i,t)}$	−1.105	−2.109**	−1.331	−2.335**
$electric_{(i,t)}$	−1.841	−2.911***	−1.203	−2.339*
$petrol_{(i,t)}$	−1.369	−2.673***	−1.914	−2.716***
$fdi_{(i,t)}$	−1.213	−2.436**	−1.141	−2.989***
$labfr_{(i,t)}$	−1.642	−2.022*	−1.247	−2.861***

Source: Authors' computation

confirming that all variables are non-stationary at level I (0) and stationary at the first difference, at various significance levels.

- Cointegration Test analysis

stat	Value	Zstat	*p*-value
Gt	−5.196	−6.996	0.000
Ga	−18.566	−2.731	0.003
Pt	−12.793	−7.245	0.000
Pa	−22.211	−4.923	0.000

co-integration analysis using the method of Westerlund (2007), showing long-run results. The lower p-value indicates a long-run association among the variables in this study.

- ARDL

Table 7 Results of cross-sectional ARDL

Long-run analysis		Short-run analysis	
Variables	Coefficients (Std error) [*p*-value]	Variables	Coefficients (Std error) [*p*-value]
$coal_{(i,t)}$	0.2500* (0.1343) [0.0626]	$coal_{(i,t)}$	0.6636* (0.3487) [0.0604]
$gas_{(i,t)}$	0.0791** (0.0308) [0.0121]	$gas_{(i,t)}$	0.2716* (0.1402) [0.0560]
$petrol_{(i,t)}$	0.1942** (0.0918) [0.0374]	$petrol_{(i,t)}$	0.2992*** (0.0881) [0.0010]
$electric_{(i,t)}$	0.1183*(0.0621) [0.0603]	$electric_{(i,t)}$	0.2331*** (0.0884) [0.0099]
$fdi_{(i,t)}$	0.01072*** (0.001) [0.0014]	$fdi_{(i,t)}$	0.0338* (0.0312) [0.01960]
$labfr_{(i,t)}$	0.4006** (0.1957) [0.0321]	$labfr_{(i,t)}$	0.0997* (0.0111) [0.0888]
		$ECM_{(-1)}$	−0.6640*** (0.1124) [0.0002]
N	140	*F*	17.9583

Note: Standard errors are given in parentheses, and $*p<0.1$, $**p<0.05$, and $***p<0.01$